

(1) Strategy (Measurement plan)

The setup for actual probing features can be changed.

(2) Alignment

The alignment system evaluating the feature can be selected.

(3) Tolerance For

A characteristics to be registered as an inspection item is selected.

(4) Nominal

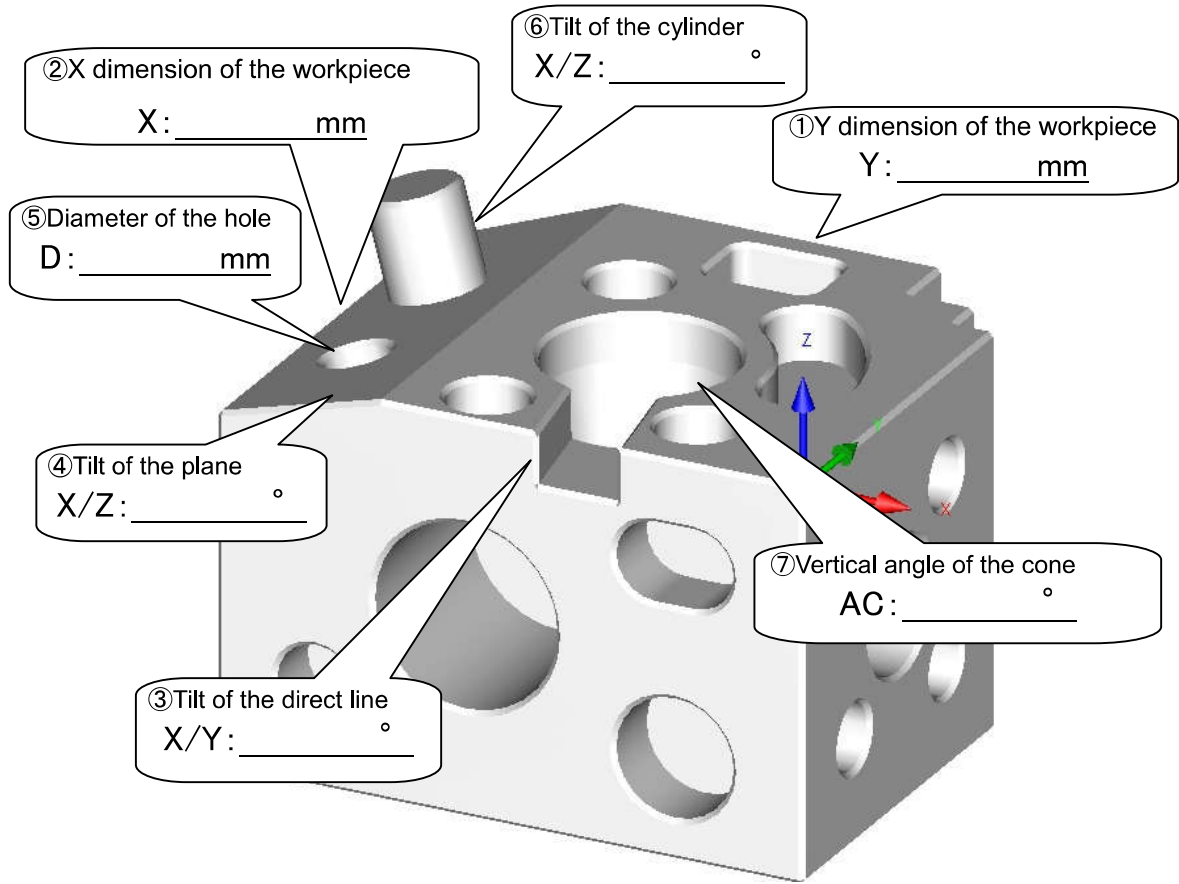
The value on a drawing.

(5) Actual

The value actually measured.

☆Let's measure!☆

Let us measure the Test block to obtain respective dimension.



[Procedure]

- (1) Y dimension of the workpiece

Measure the point and obtain the Y dimension of the workpiece from the Y value.

- (2) X dimension of the workpiece

Measure the point and obtain the X dimension of the workpiece from the X value.

- (3) Tilt of the direct line

Measure the 2D direct line and obtain the groove angle.

- (4) Tilt of the slanted plane

Measure the plane to obtain the angle of the slanted plane.

- (5) Diameter of the hole

Measure the hole opened in a slanted plane to obtain its diameter.

- (6) Tilt of the cylinder

Measure the cylinder to obtain tilted angle.

- (7) Open angle of the cone

Measure the cone to obtain its vertex angle.

[Reference values] ①: 75.0mm, ②: 90.0mm, ③: -10°, ④: -15°, ⑤: 10.0mm, ⑥: -15°, ⑦: 15°

- (1) Y dimension of the workpiece: Measure the point and obtain the Y dimension of the workpiece from the Y value.

Features		
Point2		
Comment		Strategy
Evaluation...		
Clearance Group	Nominal Definition	Alignment
CP +Y	Options	(Base Alignment)
Tolerance For:		
☐ X	Nominal	Actual
☒ Y	75.0000	75.0000
☐ Z	-7.8000	-7.8413
A1 Z/Y	0.0000	0.0000
A2 X/Y	0.0000	0.0000
Space Axis	Y	Y
☐ PtDist		0.0414

- (2) X dimension of the workpiece: Measure the point and obtain the X dimension of the workpiece from the X value.

Features		
Point3		
Comment		Strategy
Evaluation...		
Clearance Group	Nominal Definition	Alignment
CP -X	Options	(Base Alignment)
Tolerance For:		
☐ X	Nominal	Actual
☒ Y	14.7000	14.6664
☐ Z	-16.9000	-16.8819
A1 Z/-X	0.0000	0.0000
A2 Y/-X	0.0000	0.0000
Space Axis	-X	-X
☐ PtDist		-0.0382

- (3) Tilt of the direct line: Measure the 2D direct line and obtain the groove angle.

Features		
2d Line2		
Comment		Strategy
Evaluation...		
Clearance Group	Nominal Definition	Alignment
CP +Z	Options	(Base Alignment)
Tolerance For:		
☐ X	Nominal	Actual
☐ Y	2.3000	2.2943
☐ Z	-3.1000	-3.1023
☐ A1 Z/Y	0.2031	0.2236
☒ A2 X/Y	-10.0000	-10.0000
Space Axis	Y	Y
Depth	12.9000	12.9389
Norm angle	90.0175	90.0193

- (4) Tilt of the slanted plane: Measure the plane to obtain the angle of the slanted plane.

Tolerance For:	Nominal	Actual
☐ X	-88.5000	-88.4884
☐ Y	4.0000	4.0428
☐ Z	-6.3000	-6.2937
☒ A1 X/Z	-15.0000	-15.0000
☐ A2 Y/Z	0.0000	-0.0000

Space Axis: $\pm$ Z

Length 1: 21.7000, Length 2: 64.8000, Start Angle: 0.0000

- (5) Diameter of the hole: Measure the hole opened in a slanted plane to obtain its diameter.

⇒ Measurement of the hole opened in a slanted plane: P50

Relative Measurement Dialog:

To: Working Plane, Meas. Ref., Bore Missing

Measured Planes: Plane1, Plane2

Working Plane: Plane2

Features: ☒ Calculate d1, ☐ Distance d1 (3.0000), ☐ Nominal Position

Evaluation: ☒ Calculate d3, ☐ Distance d3 (0.0000), ☐ Actual position

Features Dialog for Circle1:

Circle, Projection: Plane2, Strategy: Evaluation...

Clearance Group: CP +Z, Nominal Definition: Actual To Nominal, Alignment: (Base Alignment)

Tolerance For:	Nominal	Actual
☐ X	-78.5000	-78.5230
☐ Y	20.0000	20.0000
☐ Z	-3.6000	-3.6235
☒ D	10.0000	10.0000
A1 X/Z	-15.0000	-15.0000
A2 Y/Z	0.0000	-0.0000

Space Axis: $\pm$ Z

Depth: 0.0000, Start Angle: 0.0000, Angle Segment: 360.0000

- (6) Tilt of the cylinder: Measure the cylinder to obtain tilted angle.

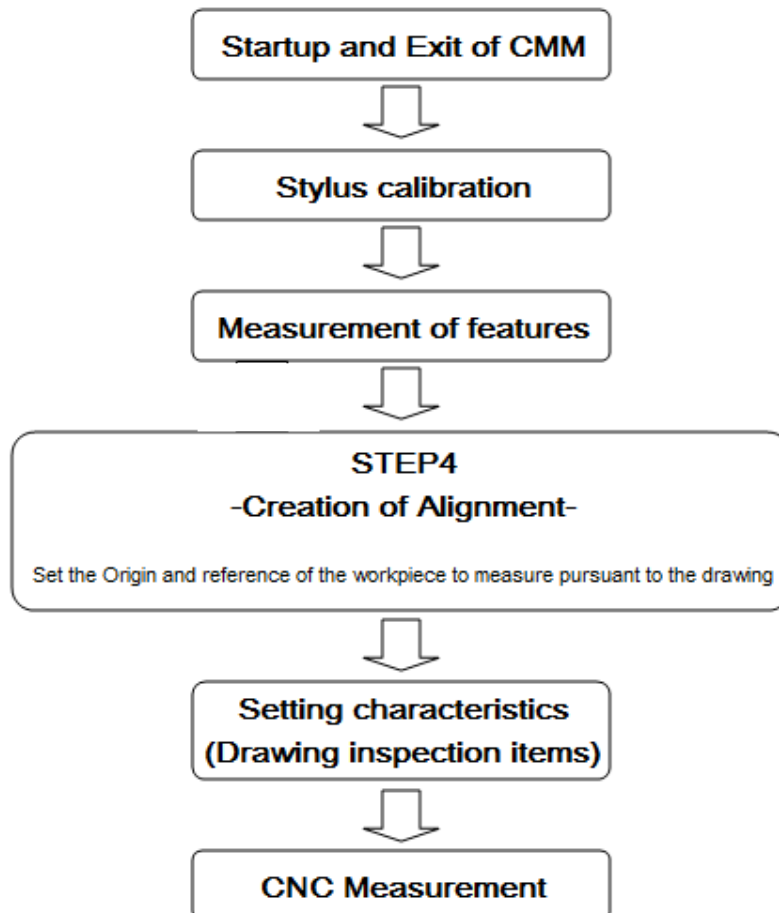
Tolerance For:		Nominal	Actual
☐ X		-79.1000	-79.1183
☐ Y		45.0000	45.0000
☐ Z		-1.4000	-1.4016
☐ D		14.0000	14.0000
☒ A1 X/Z		-15.0000	-15.0000
☐ A2 Y/Z		0.0000	0.0000
Space Axis	Z		Z
Depth		10.7000	10.7020
Start Angle		0.0000	0.0000
Angle Segment		360.0000	360.0000

- (7) Open angle of the cone: Measure the cone to obtain its vertex angle.

Tolerance For:		Nominal	Actual
☐ X		-40.0000	-40.0000
☐ Y		30.0000	30.0000
☐ Z		-27.3000	-27.2899
☐ D		22.5000	22.5408
☐ A1 X/Z		0.0000	-0.0000
☐ A2 Y/Z		0.0000	0.0000
Space Axis	Z		Z
☒ Cone Angle		15.3000	15.3239
Depth		18.2000	18.2410
Start Angle		0.0000	0.0000

<< Memo >>

Creation of Alignment

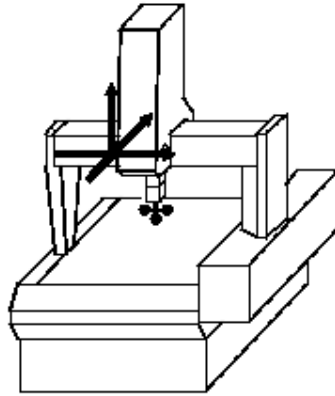


6-1 Alignment

For doing measurement, the alignment (reference) can be broadly divided into two.

A. Machine alignment

This is the alignment to be specified when the power is turned on, which is specified in parallel to each axis of the measurement machine.

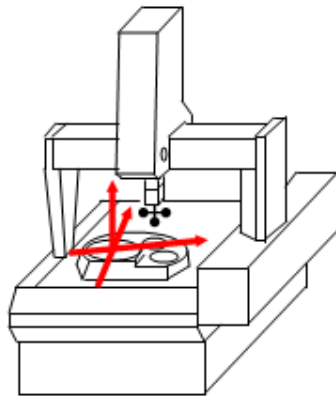


Machine alignment

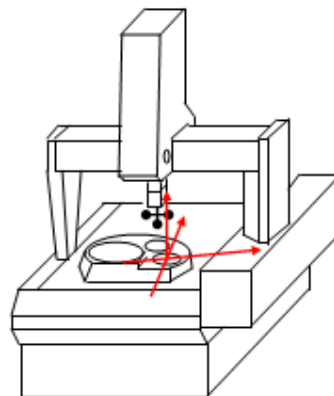
B. Work alignment (base alignment and evaluation alignment)

This is the Alignment to be specified on measurement works. Normally, all the measurement data is outputted from this Alignment.

The base alignment is the base of measurement. If it is difficult to evaluate with the base alignment, an evaluation alignment can be specified in order to evaluate the part alone.



Base alignment



Evaluation alignment

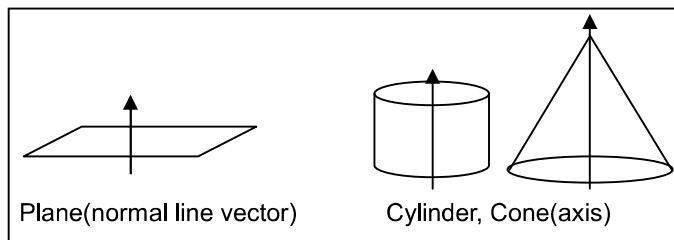
(1) Purpose of creating alignment

In terms of location, inclination, and direction, the setup details of a measurement object placed on a surface plate of CMM is unknown. Therefore, alignment is specified in order to teach that information.

(2) Concept of alignment setup

Spatial Rotation

At the beginning, the inclination of whole work is taught.
To teach the inclination, a three-dimensional feature inclined is used.
The features available are shown as follows:



Planar Rotation

Next, the direction to which the work is facing is taught.

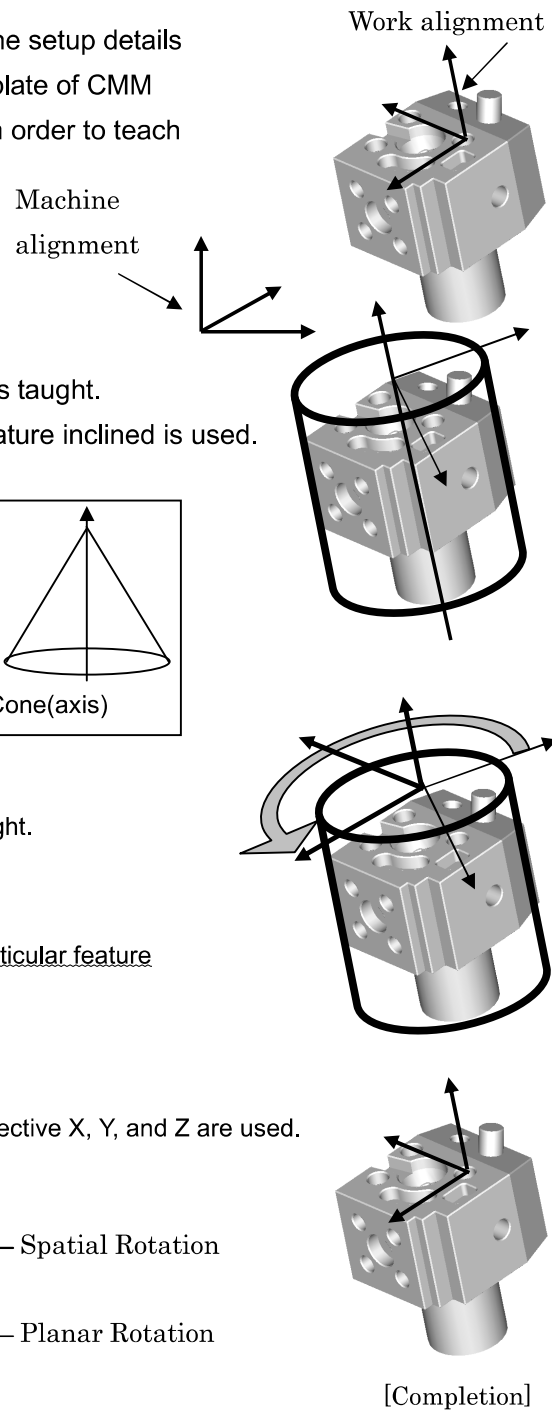
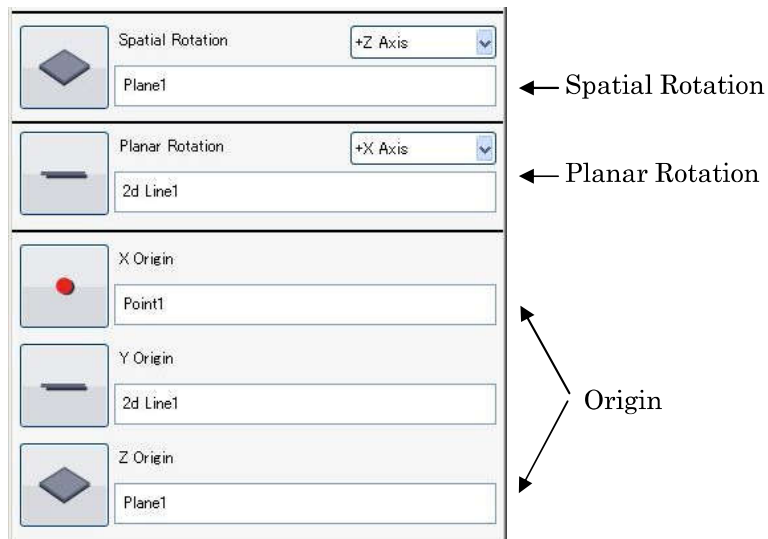
There are two methods for teaching the direction.

1. Using two-dimensionally inclined features
2. Letting the work face the direction of a particular feature from the origin

Origin

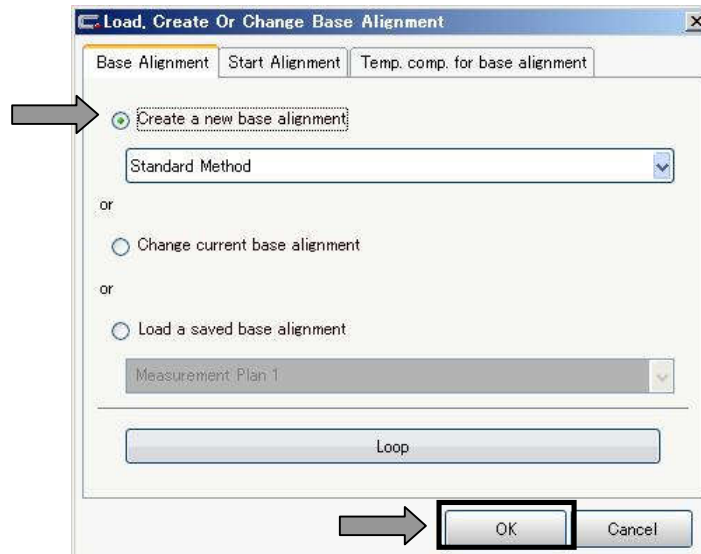
Finally, the position of a work is taught.

A feature possessing the alignment effective to respective X, Y, and Z are used.



6-2 Alignment setup

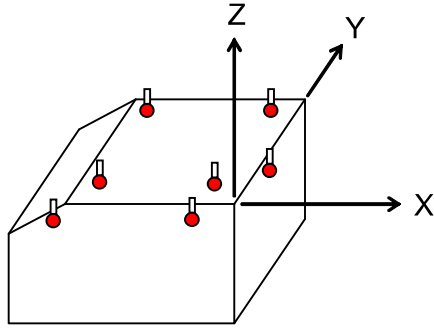
(1) Click “Base/Start Alignment” icon , select “Create a new base alignment”, and click “OK”.



(2) Base Alignment template is displayed.

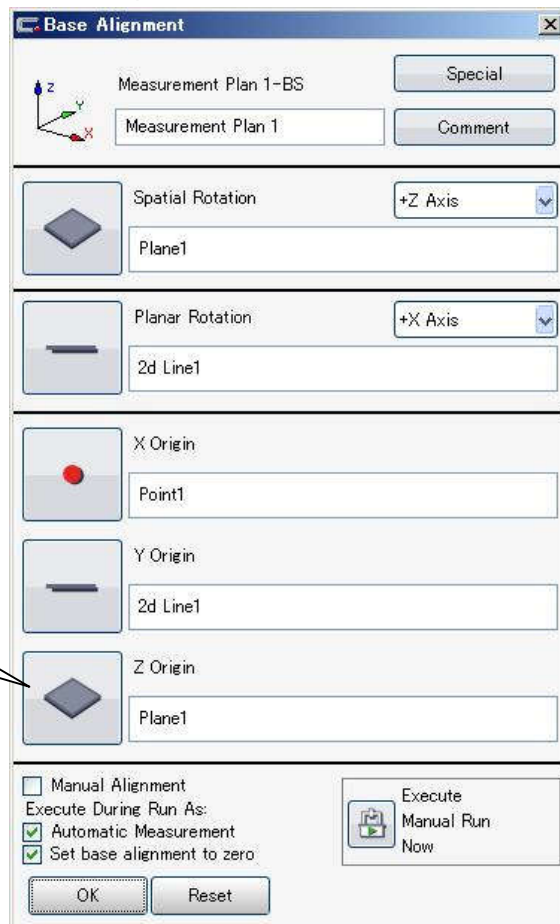


Example – For specifying alignment at the front right angle on the upper surface of a work as the figure below



- Perform probing for three or more points on the upper surface, and click “OK” to finish this step.
(It is recognized as a plane by the AI function)
- Perform probing for two or more points on the front surface, and click “OK” to finish this step.
(It is recognized as a straight line by the AI function)
- Perform probing for one point on the right side surface, and click “OK” to finish this step.
(It is recognized as a point by the AI function)

As a result of the above operation, three features including Plane1, 2-D Line1, and Point1 are created and an alignment for the work is specified



When a measurement is performed with the template opened, it is allocated to the alignment automatically, however, it also can be allocated manually by clicking the square icon.

When a measurement is proceeded while opening the base alignment template as above, Calypso automatically specifies a alignment within the basic alignment template using the features applicable to the datum field. To specify the basic alignment manually, measure each feature in advance, open the base alignment template, and allocate the features to the respective datum.

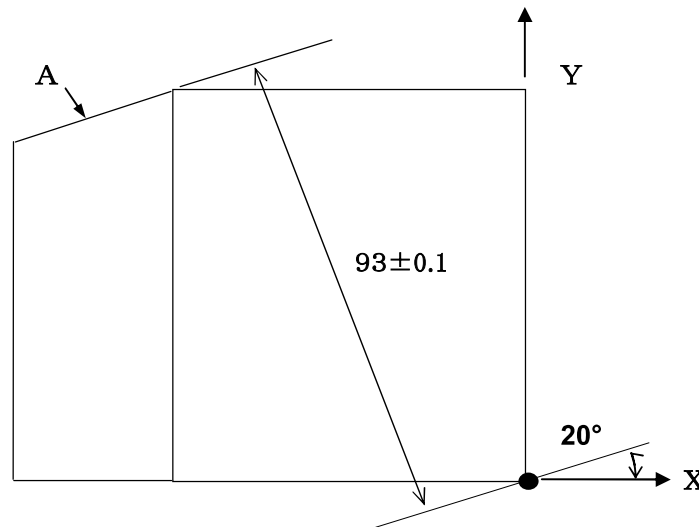
(3) Click "OK" below the template to go back to the User Desk.

6-3 Evaluation alignment

To change the alignment while a measurement is on the way, the base alignment is not changed depending on the measurement location.

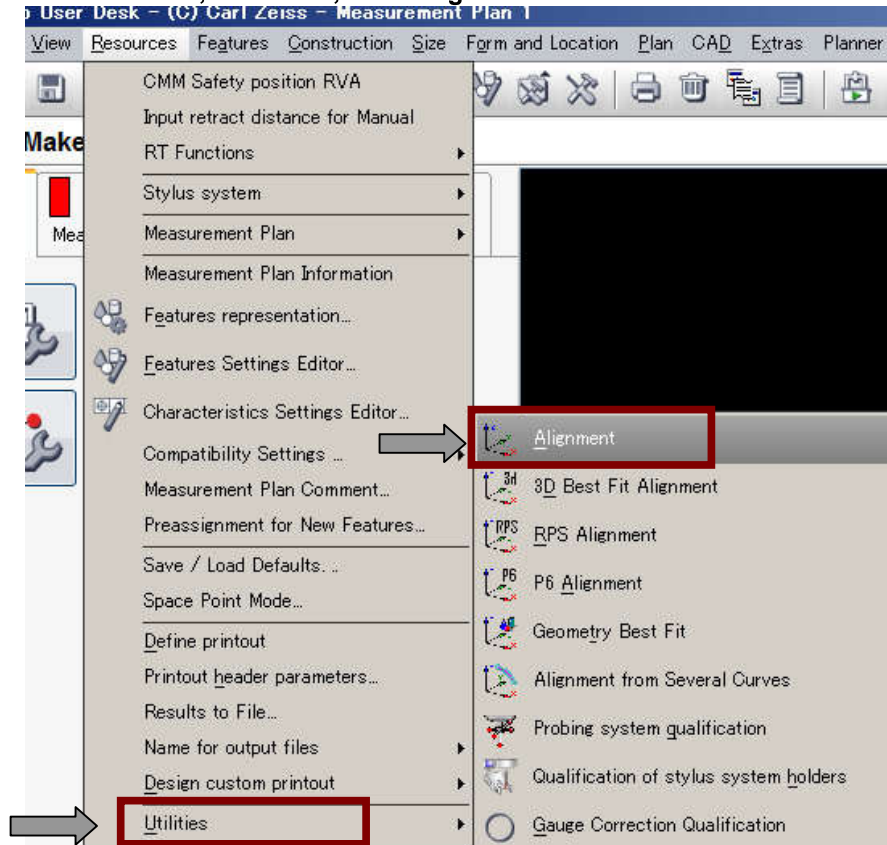
An example is shown below.

Calculation of the distance to the Edge A (93 plus or minus 0.1)

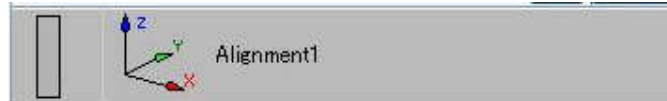


- (1) It is assumed that the above base alignment is already specified.
- (2) In order to measure the "93 plus or minus 0.1", a alignment is added.

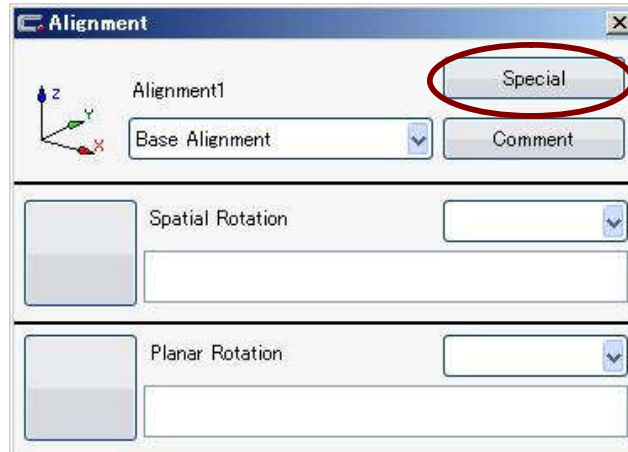
Select **"Resources", "Utilities", and "Alignment."**



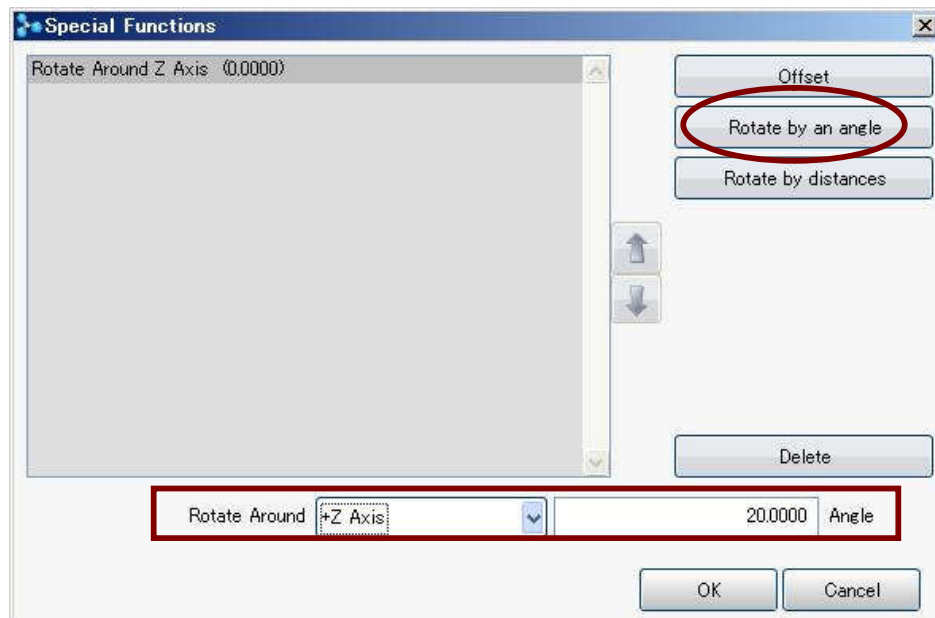
(3) Alignment1 is created in the Characteristics list.



(4) Open Alignment1 icon and click "Special" button.



(5) Next, click "Rotate by an angle" and input the value as "Rotate Around: +Z" and "Angle: 20."

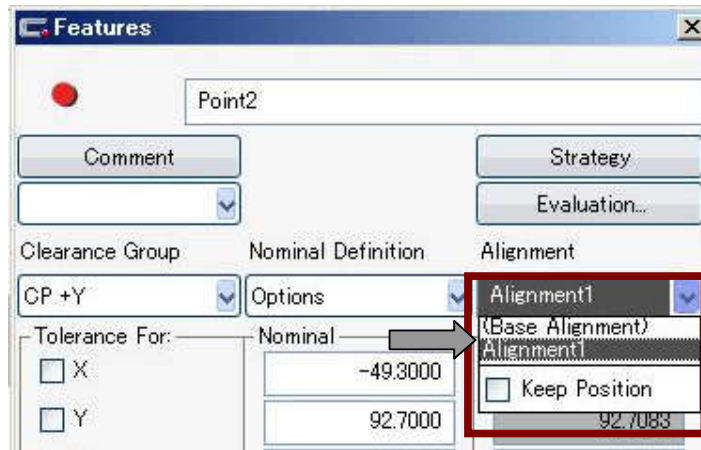


(6) After the input, click "OK" to close the Alignment1 setup.

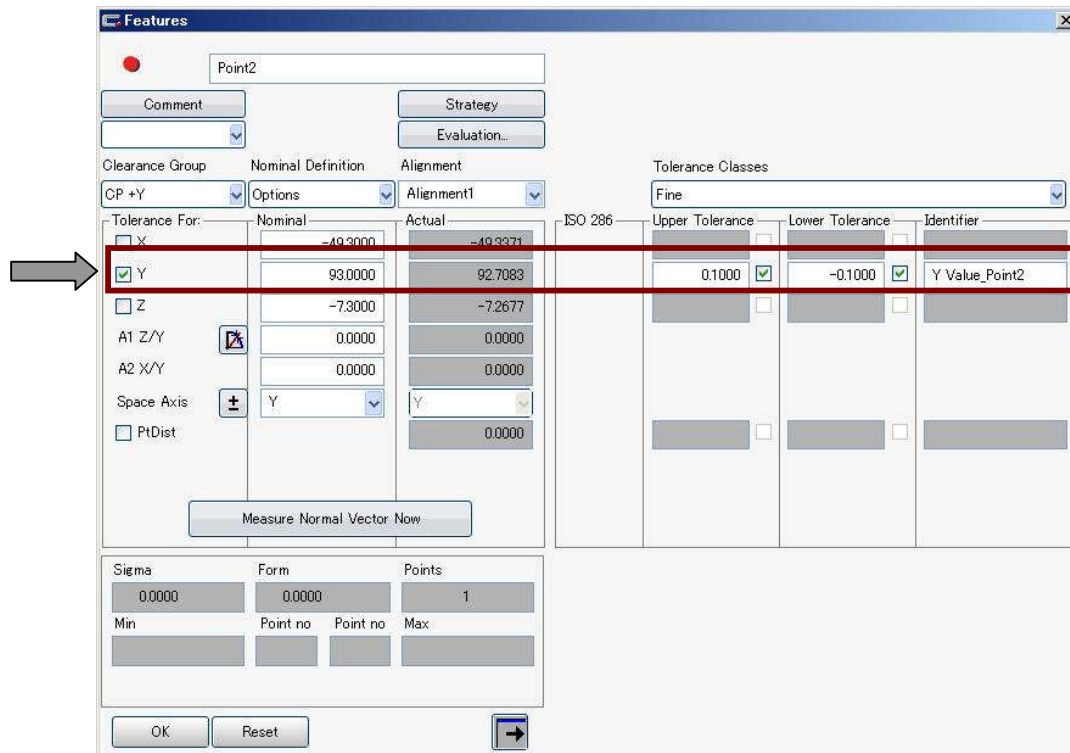
(7) Next, measure a point on the Edge A. (point measurement)

After the measurement, perform the following setup before exiting.

Click ▼ of the "Alignment" to change the menu from "Base Alignment" to "Alignment1."



(8) Input the Nominal value and the Tolerance for Y, and click "OK."



6-4 Example of alignment setup

(1) Plane reference

Two holes are the reference.

Spatial Rotation	+Z Axis
PlaneA	
Planar Rotation	+X Axis
CircleC	
X Origin	CircleB
Y Origin	CircleB
Z Origin	PlaneA

Hole and edge plane are the reference.

Spatial Rotation	+Z Axis
PlaneA	
Planar Rotation	+Y Axis
2d LineC	
X Origin	CircleB
Y Origin	CircleB
Z Origin	PlaneA

Reference hole is at the center.

Spatial Rotation	+Z Axis
PlaneA	
Planar Rotation	+X Axis
Symmetry1	
X Origin	CircleB
Y Origin	CircleB
Z Origin	PlaneA

The hole is offset (1 direction)

Spatial Rotation	+Z Axis
PlaneA	
Planar Rotation	+X Axis
CircleC	
X Origin	CircleB
Y Origin	CircleB
Z Origin	PlaneA

The hole is offset (2 directions)

Spatial Rotation	+Z Axis
PlaneA	
Planar Rotation	+X Axis
CircleC	
X Origin	CircleB
Y Origin	CircleB
Z Origin	PlaneA

Special

1. Reference axis setting in secondary reference

☒ Rotate to distance to the +X axis: 0.0000

☐ Rotate to origin line: 0.0000 X 0.0000 Y

☐ Rotate to same deviation: 0.0000 X 0.0000 Y

Special

1. Reference axis setting in secondary reference

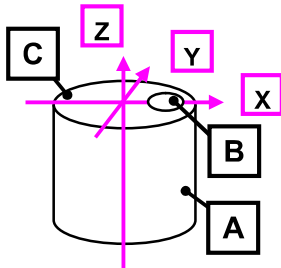
☐ Rotate to distance to the +X axis: 0.0000

☒ Rotate to origin line: 0.0000 X 0.0000 Y

☐ Rotate to same deviation: 0.0000 X 0.0000 Y

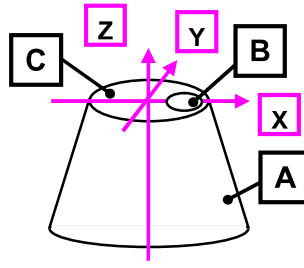
(2) Cylinder reference

One cylinder axis and knock hole is the reference.



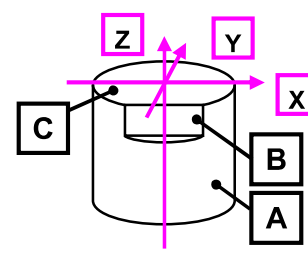
	Spatial Rotation	+Z Axis
	Planar Rotation	+X Axis
	X Origin	CylinderA
	Y Origin	CylinderA
	Z Origin	PointC

One taper axis and knock hole is the reference.



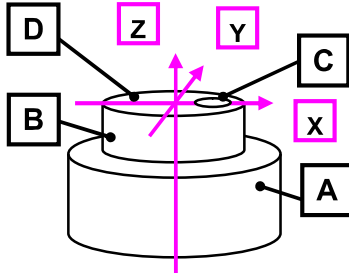
	Spatial Rotation	+Z Axis
	Planar Rotation	+X Axis
	X Origin	ConeA
	Y Origin	ConeA
	Z Origin	PointC

One cylinder axis and edge face is the reference.



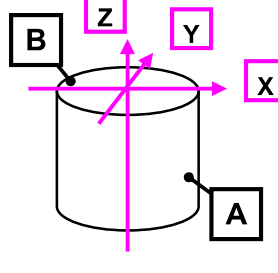
	Spatial Rotation	+Z Axis
	Planar Rotation	+X Axis
	X Origin	CylinderA
	Y Origin	CylinderA
	Z Origin	PointC

Axis of two cylinders and knock hole is the reference.



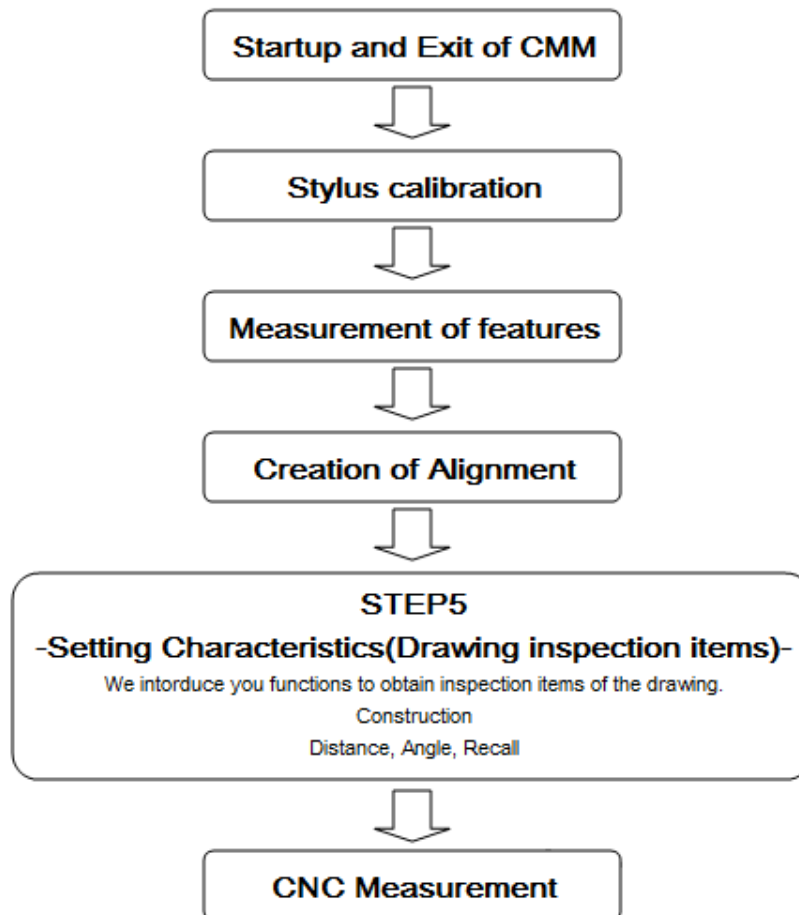
	Spatial Rotation	+Z Axis
	Planar Rotation	+X Axis
	X Origin	3d Line
	Y Origin	3d Line
	Z Origin	PointD

Cylinder axis without rotation reference.



	Spatial Rotation	+Z Axis
	Planar Rotation	+X Axis
	X Origin	CylinderA
	Y Origin	CylinderA
	Z Origin	PointB

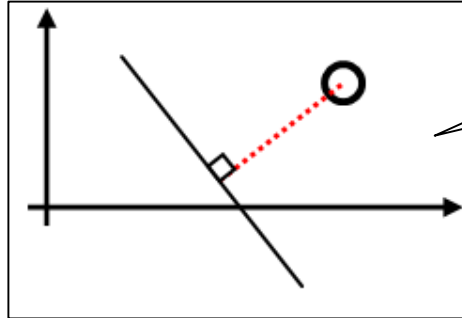
Inter-feature calculation



7-1 Constuction

- (1) **Perpendicular** ... A vertical line is drawn between two features.

"Construction" ⇒ "Perpendicular"



A vertical line is drawn downward from the first feature to the second feature.

Perpendicular

Perpendicular1 Comment

Alignment (Base Alignment) ▼

Feature 1
Circle1

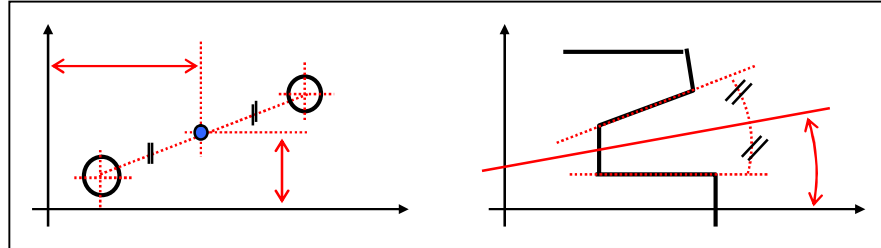
Feature 2
2d Line2

Tolerance For:	Nominal	Actual
☐ X	-18.0313	-17.9983
☐ Y	-51.2827	-51.2548
☐ Z	6.4000	6.3888
☐ A1 Z/Y	0.0000	0.0000
☐ A2 X/Y	40.0000	39.9839
Space Axis	Y	Y
☐ Depth	66.9448	66.9064
Start Angle	0.0000	0.0000

OK Reset

- (2) **Symmetry** ...The symmetry (center) data is calculated based on two features. That is, if two circles are allocated, the center of the circles is defined, and if two straight lines are allocated, the central line is defined.

"Construction" ⇒ "Symmetry"



Symmetry

Symmetry1 Comment

Alignment (Base Alignment) ▼

Feature 1
Circle1

Feature 2
Circle2

Result Selection
1 2

Tolerance For:	Nominal	Actual
☐ X	6.2500	6.2469
☐ Y	-10.8500	-10.8200
☐ Z	6.6500	6.6419

OK Reset ➔

(3) **Intersection** ... The intersection data is calculated based on two features.

For an intersection between a cylinder and a plane as the figure below, it is possible to select the intersection between a plane and a cylindrical axis, or the circle formed by slicing a cylinder flatly, in addition to an intersection between lines.

"Construction" ⇒ "Intersection"

